QIRI (Quantum Integrated Research Institute Inc.)
Tokyo 107-0061, Japan
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Editors, *Physical Review B*

Dear Editors,

We submit the manuscript "Exact exponential heat rectification from the Liouvillian skin effect" by Hikaru Wakaura and Taiki Tanimae for consideration as a regular article in Physical Review B.

Summary. We study heat transport through a reciprocal tight-binding chain with directional bond dissipation between two thermal leads. The ratio of the forward and reverse heat-current responses obeys an exact law, $\delta Q_f/\delta Q_r = \exp[2(L-1)/\xi]$, with ξ the Liouvillian skin localisation length. The law follows from an identity of tridiagonal Green functions and holds at every energy and temperature and for arbitrary on-site disorder. The manuscript then maps its domain of validity: (i) with next-nearest-neighbour hopping the exponent is the energy-resolved non-Bloch localisation length, although the spectral winding number vanishes at all transport energies; (ii) superfermion matrix-product-state simulations, calibrated against the exact solution, show that a nearest-neighbour interaction up to $2J$ leaves ξ unrenormalised; (iii) a finite-temperature bulk bath closes the entropy balance, with a cost per rectified heat that grows as $1/\xi$; (iv) bond disorder obeys an additive sum rule.

Why Physical Review B. The non-Hermitian skin effect, non-Bloch band theory and their transport signatures have been developed to a large extent in this journal, and our work addresses that readership directly: it connects the Liouvillian skin effect to a measurable two-terminal thermal response, identifies which bulk quantity controls it, and tests the result against interactions and disorder with controlled numerics. Recent related work in Physical Review B includes Liouvillian skin effects at finite temperature [Phys. Rev. B **113**, 035121 (2026)] and circuit observations of skin effects [Phys. Rev. B **111**, 224104 (2025)]. All derivations, the precision analysis of the numerics and the experimental parameter estimates are contained in the main text.

Previous submission. A shorter version of this work was submitted to Physical Review Letters as a Letter (manuscript LX20531) and was not sent for external review; the editors stated that they had not judged the validity of the manuscript. The present manuscript is a full-length article in which the material formerly in the Supplemental Material is integrated into the main text. It has not been submitted to any other journal.

Suggested referees (no conflict of interest with the authors):

- Zhong Wang, Tsinghua University (non-Hermitian skin effect, chiral damping)
- Tsuneya Yoshida, Kyoto University (Liouvillian skin effects)
- Kazuki Yokomizo, University of Tokyo (non-Bloch band theory)
- Henning Schomerus, Lancaster University (transport in non-Hermitian systems)
- Dario Poletti, Singapore University of Technology and Design (boundary-driven open quantum systems)

Data. All data and the code that regenerate every figure are deposited at Zenodo (doi:10.5281/zenodo.23801926) and on GitHub (https://github.com/deeptell-inc/skinheat). A preprint will be posted on arXiv after publication.

Sincerely,

Hikaru Wakaura and Taiki Tanimae